

Theo Heron

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Education

Columbia University · New York, NY

Expected May 2028

M.S. in Chemical Engineering

UC San Diego · La Jolla, CA

Expected June 2026

B.Sc. in Physics | Minor in Chemistry

Relevant courses: Electromagnetism I, II & III, Circuits Lab, Quantum Chemistry Lab

Research Experience

- **Chemical Engineering REU – UChicago (Prof. Y. Shirley Meng)** Summer 2025
 - Increased **LMFP cathode areal capacity by 33%** through optimized slurry casting thickness.
 - Fabricated and cycled **40 coin cells** in a glove box; correlated how low porosity and material degradation hampered electrochemical performance.
 - Evaluated **coulombic efficiency, capacity retention, and dQ/dV curves**; carried out analysis in **Python and MATLAB**. [\[Click for poster\]](#)
- **Undergraduate Research – UC San Diego (Prof. Oleg Shpyrko)** 2024 - Present
 - Conducted **synchrotron experiments** at Argonne National Laboratory's Advanced Photon Source to study temperature-dependent Na⁺ dynamics in solid electrolytes.
 - Working on diffraction analysis and **Rietveld refinement** to characterize electrolyte structure.
 - Organized group meetings and hosted a Journal Club on solid-state battery literature.

Publications & Presentations

- **Poster:** *Optimizing LMFP cathode areal capacity for Li-ion cells* – UChicago PME Capstone Poster Session (2025).
- **Presentations:** Presented solid-state battery papers at UCSD Physics Journal Club (2024).
- **Paper:** *Li-FeS₂ vs. Zn/MnO₂ AA Cells Energy Capacity* (IB Extended Essay, 2022). [\[Click for paper\]](#)

Leadership

- **Society of Physics Students, UC San Diego**
 - **Treasurer (2025 - Present):** Manage \$3,000 annual budget & process reimbursement requests. Successfully secured funding in spite of funding cuts.
 - **Media Team (2022 - 2024):** Organized outreach events, designed posters, managed website & discord server to bring our physics community closer together.

Skills

Experimental: Glovebox, battery cycler, hot roller, slurry casting, XPCS, oscilloscope

Programming: Python, MATLAB, Linux/Unix, HPC (SDSC Expanse), Avogadro

Music Production: Vocal recording, production, arrangement, mixing, Ableton Live, Logic Pro

Languages: English (native), French (native)